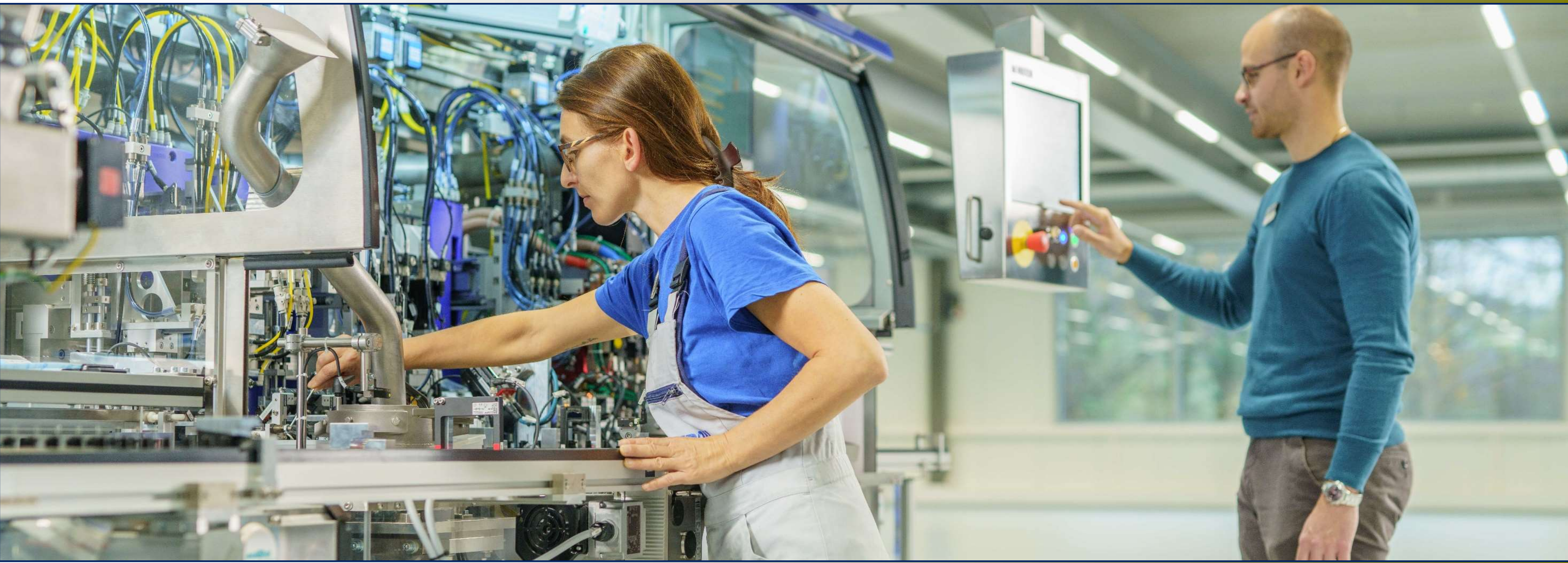
MIKRON AUTOMATION



Mikron Automation Training Instruction – G05 Education Series™

Level 0.4 – Preventive Maintenance V6.1

Foreword:

This **G05 Education Series™** presentation describes the basics of the various preventive maintenance frequencies recommended by Mikron automation for the G05. Version 6.1 G05 was referenced when creating this presentation. The standard G05 documentation hold all the info used in this presentation.

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1

Weekly Checks

Preventive Maintenance Mikron G05

Weekly Maintenance Checks

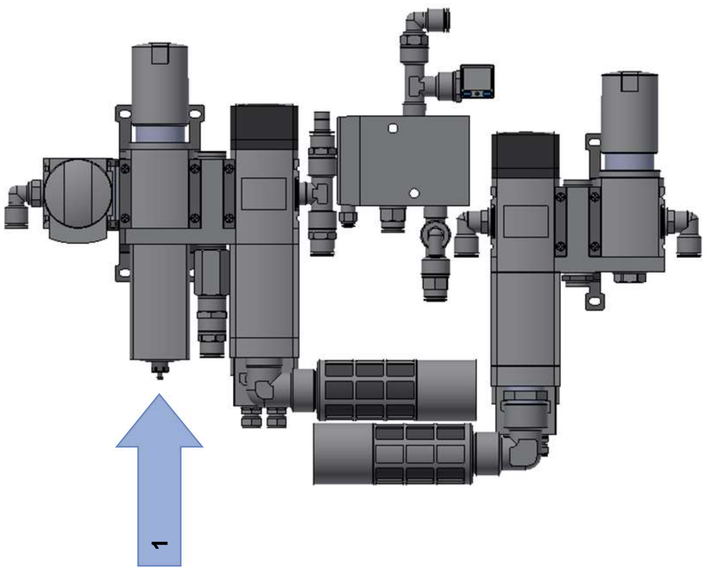
Every week the following tasks must be performed.

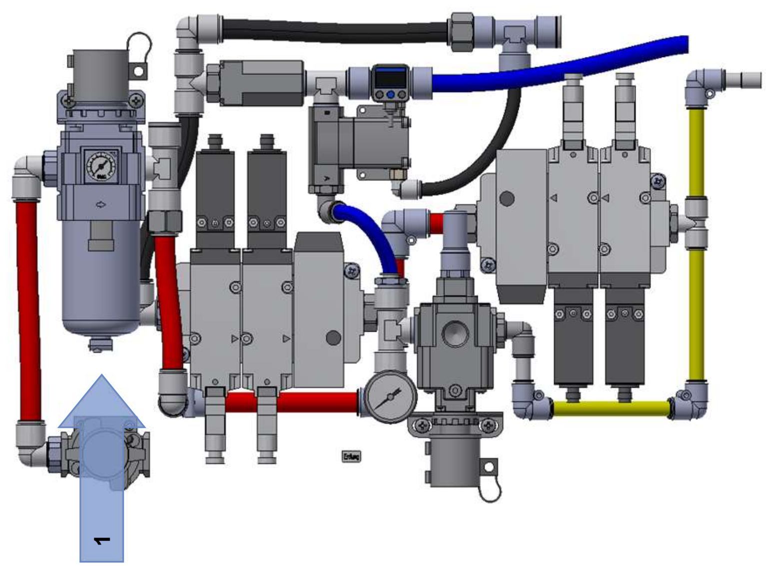
- Compressed air regulator water separator trap
- Conveyor belt system

Weekly Checks

Compressed air regulator water separator trap

- Water in the air supply may damage the assembly system.
- The water separator trap (1) should be checked weekly.

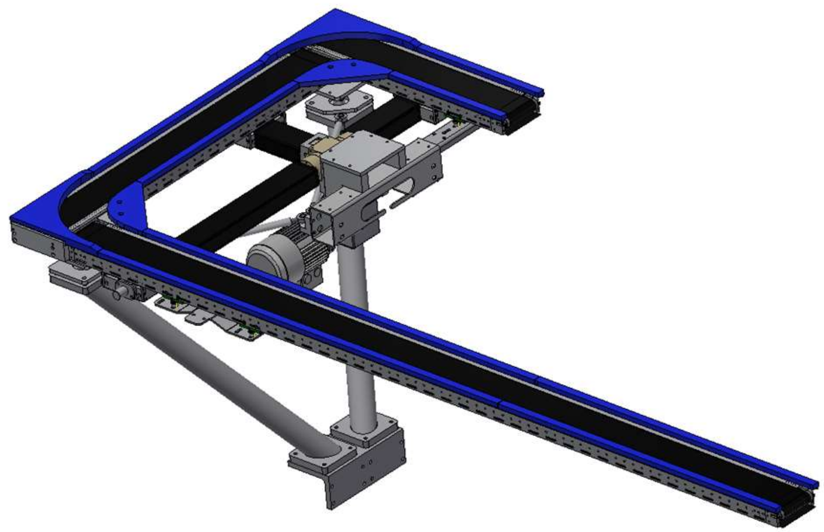




Weekly Checks

Conveyor belt system

- To minimize particulate generation it is recommended that the transport belts are regularly cleaned.
- Belts should be cleaned with a solution of isopropyl alcohol & distilled water, 70:30 ratio.
- Cleaning products containing acetone must not be used as this would damage the belt material.
- Keep the belts running in the middle of the rolls.
- Check for parts trapped in the belt corners and between conveyors.



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Quarterly (4x per year)

or 5 Million Cycles

Preventive Maintenance Mikron G05

Quarterly Maintenance (4x per year) or every 5 million cycles

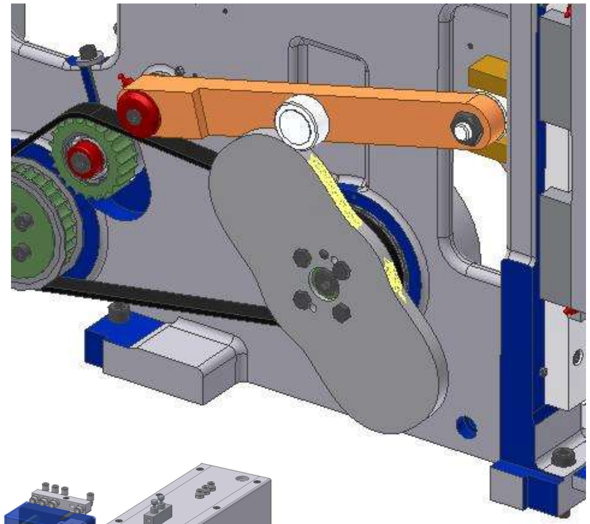
The following tasks must be performed.

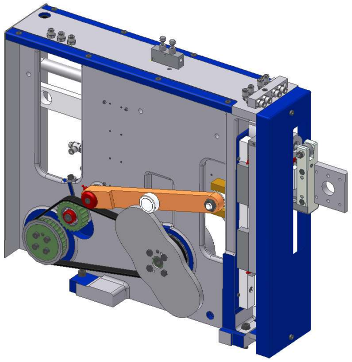
- Lubrication of all the P&P and vertical units cams & geared wheels
- Visual check of the pallets
- Cleaning of the Pallets nest (more often if required)
- Visual check of the locking pin
- Visual check of the transfer bushings
- Cleaning the V–guide and front guide
- Check of the Linear motor (Lubrication if required)
- Check & lubrication of drive cams & rollers
- Check gearbox oil level
- Lubrication of Rise and Fall’s cams & rollers
- Maintenance points on MiMoGa (if present)

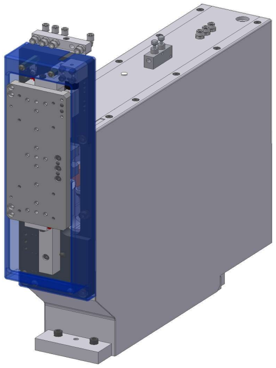
Quarterly Maintenance (4x per year) or every 5 million cycles

Lubrication

- Grease & wipe off any surplus grease on rollers (external part).
- Lubricate all the cams inside the units.
- It is important to first clean off the old grease, inspect the surface of the cam, and then apply a thin layer of grease.
- Lubricate the geared wheel.
- See the greasing chart.



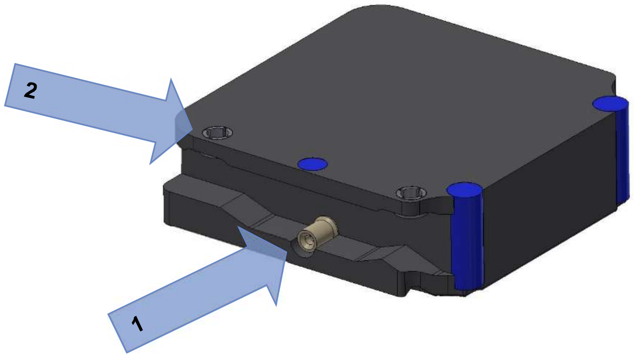


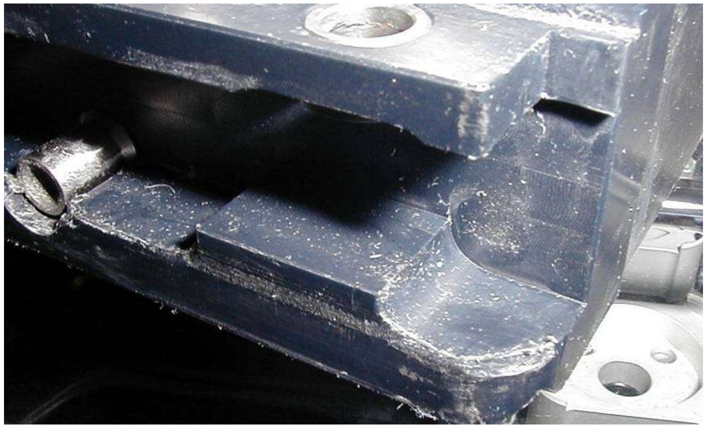


Quarterly Maintenance (4x per year) or every 5 million cycles

Pallets

- Check the underside of the pallets for damage.
- Check that the ”V” guide slot under the pallet as well as the mobile stop area do not have any signs of wear as this may contribute to poor accuracy of assembly .
- The pallet locking pin (1) and the transfer bushings (2) should be cleaned, check & greased.
- The pallets can be cleaned more often if required.



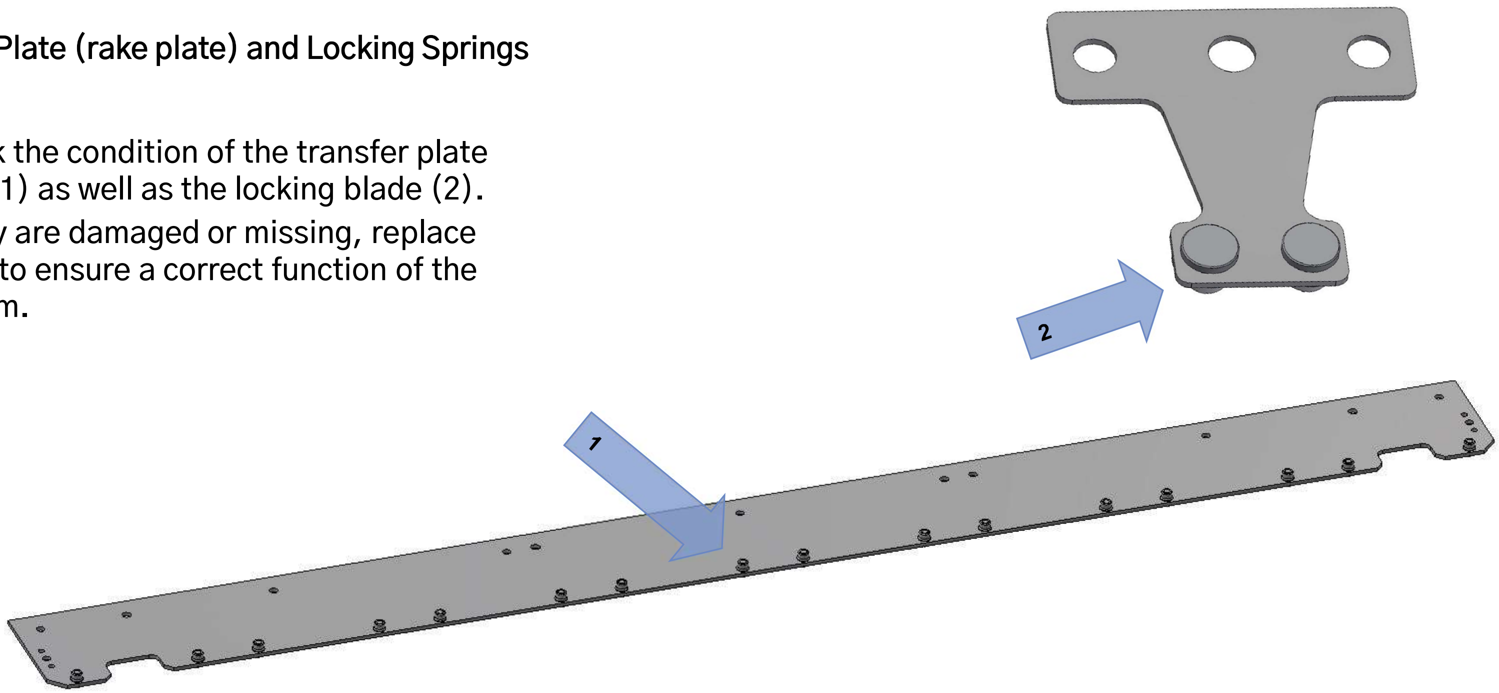


Example of a damaged pallet

Quarterly Maintenance (4x per year) or every 5 million cycles

Transfer Plate (rake plate) and Locking Springs

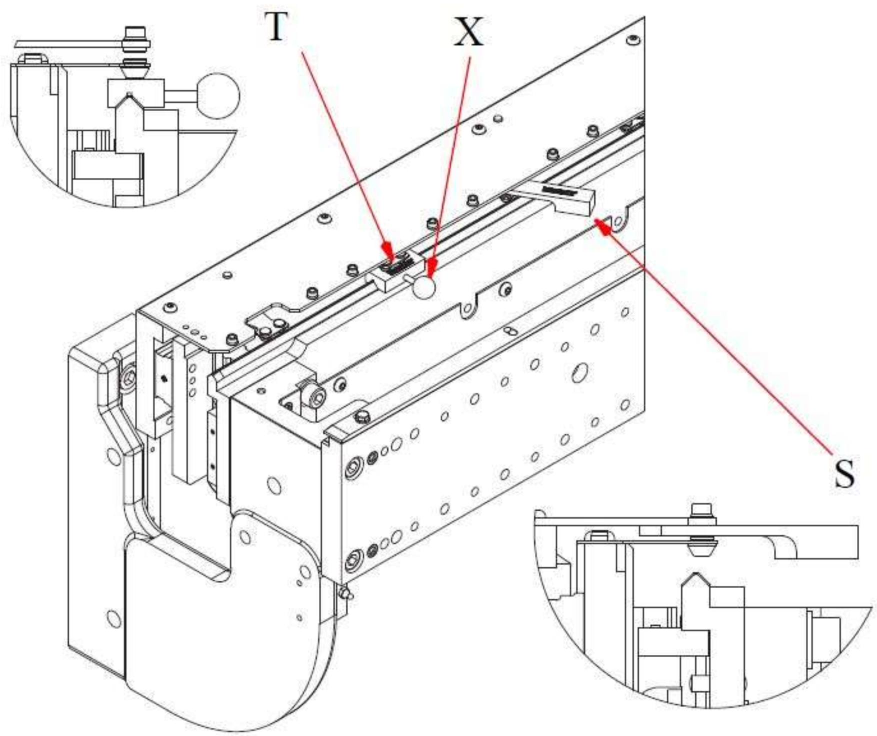
- Check the condition of the transfer plate pins (1) as well as the locking blade (2).
- If they are damaged or missing, replace them to ensure a correct function of the system.



Quarterly Maintenance (4x per year) or every 5 million cycles

Locking Spring Checks:

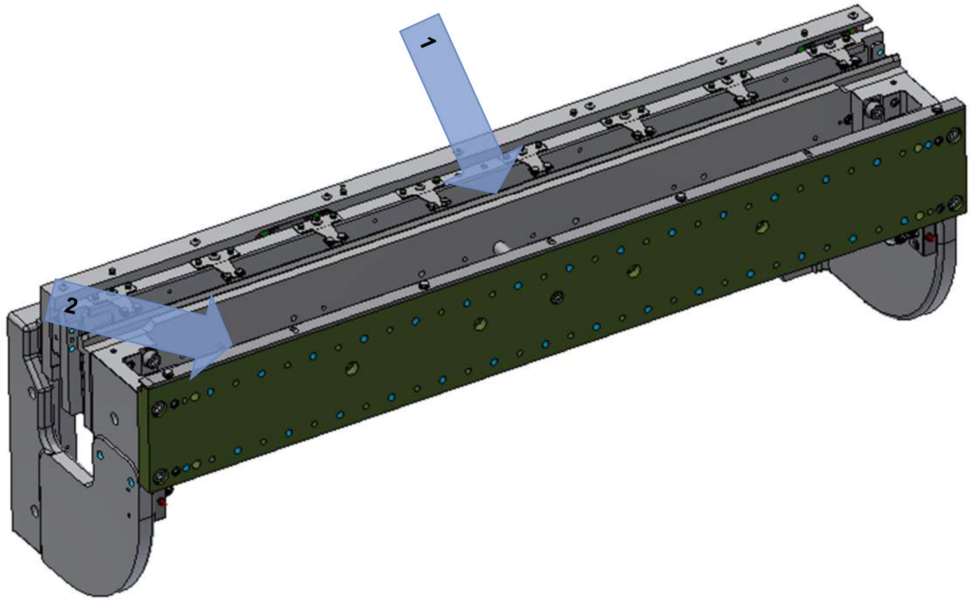
- Check of the locking system.
- The machine has to be locked in the "transfer position" (normally between 150° and 350°).
- The supplied tool S4085.0028832 is necessary to carry out this task.
- For more information refer to the standard documentation chapter 441.



Quarterly Maintenance (4x per year) or every 5 million cycles

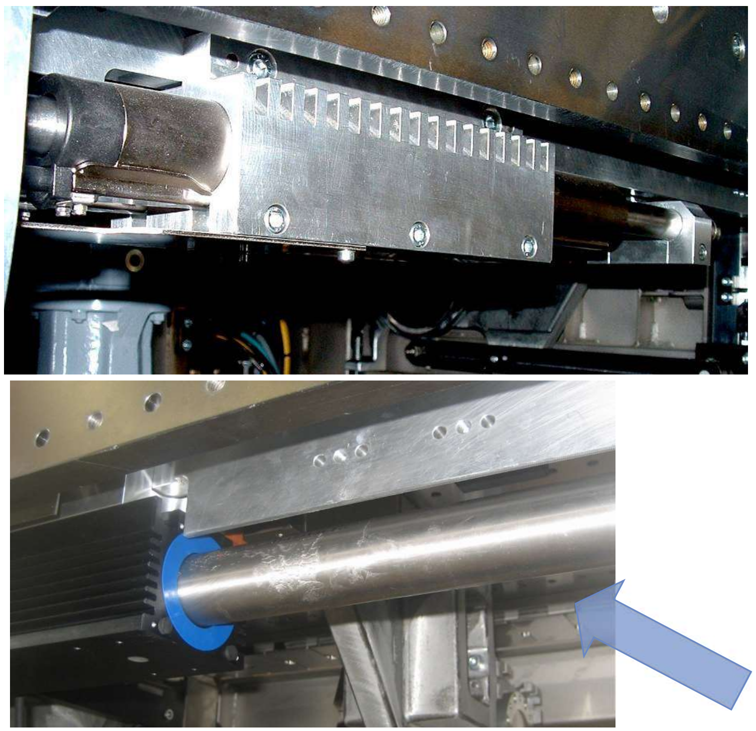
Index Unit Pallet Guides

- The machine should be emptied of pallets.
- The "V" guide (1) cleaned with isopropyl alcohol.
- The “front” guide (2) cleaned with isopropyl alcohol.



Quarterly Maintenance (4x per year) or every 5 million cycles

Linmot Checks



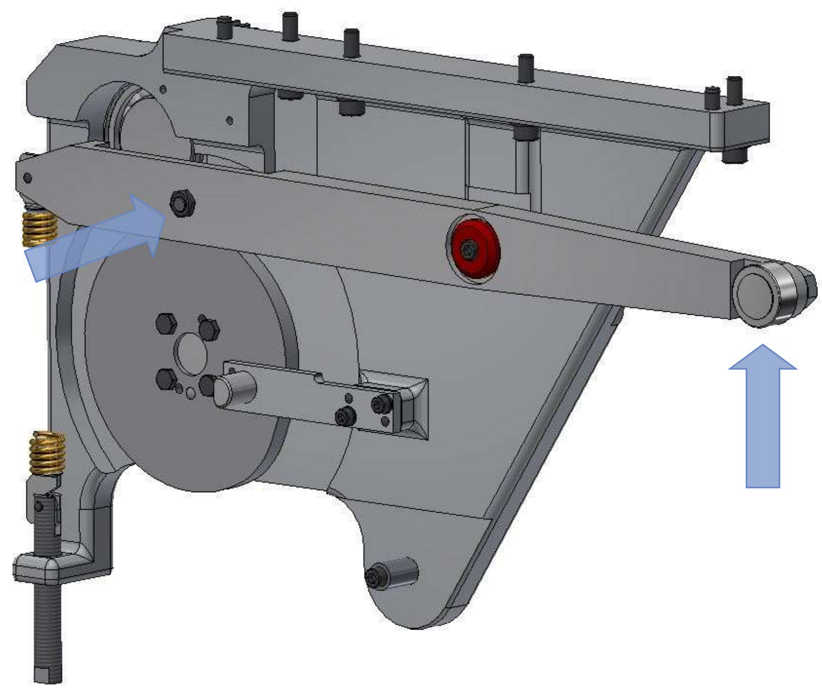
- Check the linear motor slider.
- After cleaning, grease them lightly.
- See the greasing chart.

BEWARE! The slider is strongly magnetized!

Quarterly Maintenance (4x per year) or every 5 million cycles

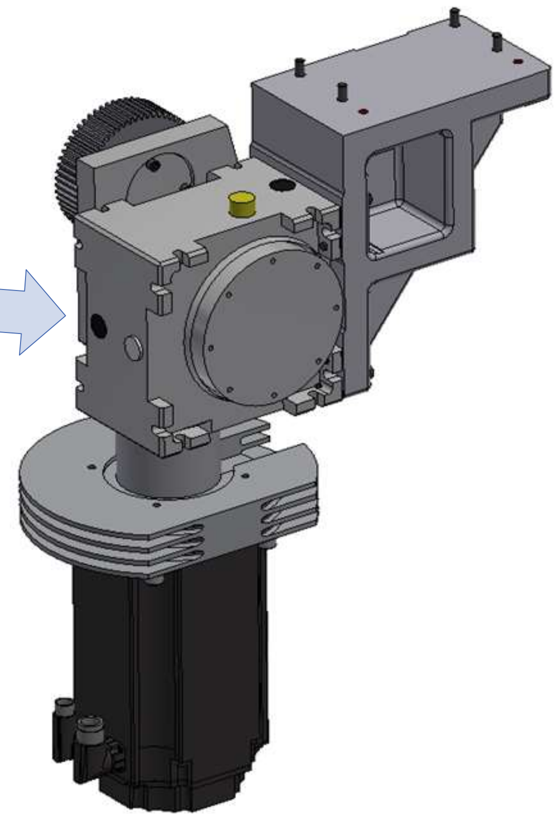
Drive Unit Lubrication

- Grease & wipe off any surplus grease on rollers (external part).
- Lubricate all the cams inside the units.
- It is important to first clean off the old grease, inspect the surface of the cam, and then apply a thin layer of grease.
- Lubricate the geared wheel.
- See the greasing chart.



Quarterly Maintenance (4x per year) or every 5 million cycles

Main Rotary Gear Box Checks



- Check the oil level (1) in the gearbox of the main drive unit. (3/4)
- If necessary adjust the oil level.
- See the greasing chart.

Quarterly Maintenance (4x per year) or every 5 million cycles

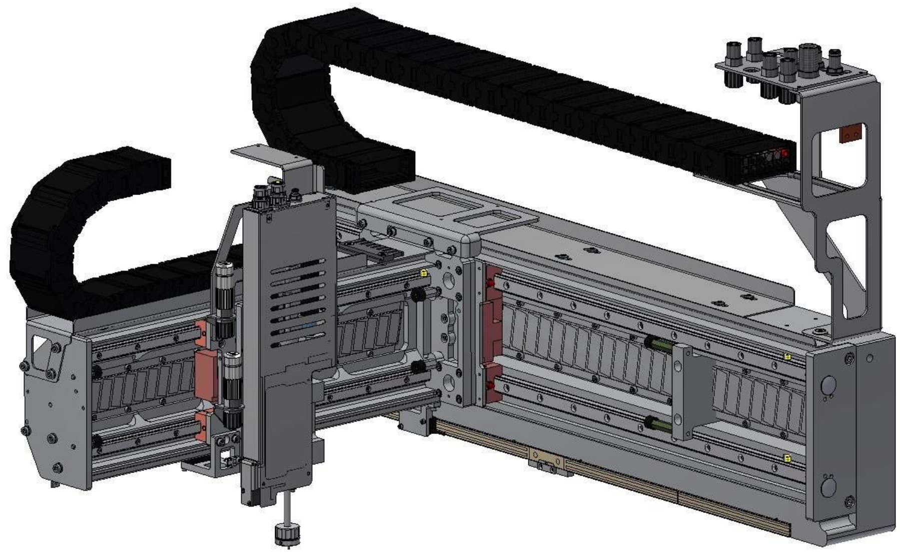
MiMoGa (Mikron Modular Gantry) Checks

If the machine is equipped with MiMoGa (Mikron Modular Gantry), refer to chapter 831 in the standard documentation.

Maintenance:

- Clean the optical scales.
- Cleaning / lubrication of all the axis.
- Inspection of cables, electrical and mechanical connections.

BEWARE! The magnetic ways are strongly magnetized!



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Biannual (2x per Year)

or 10 Million Cycles

Preventive Maintenance Mikron G05

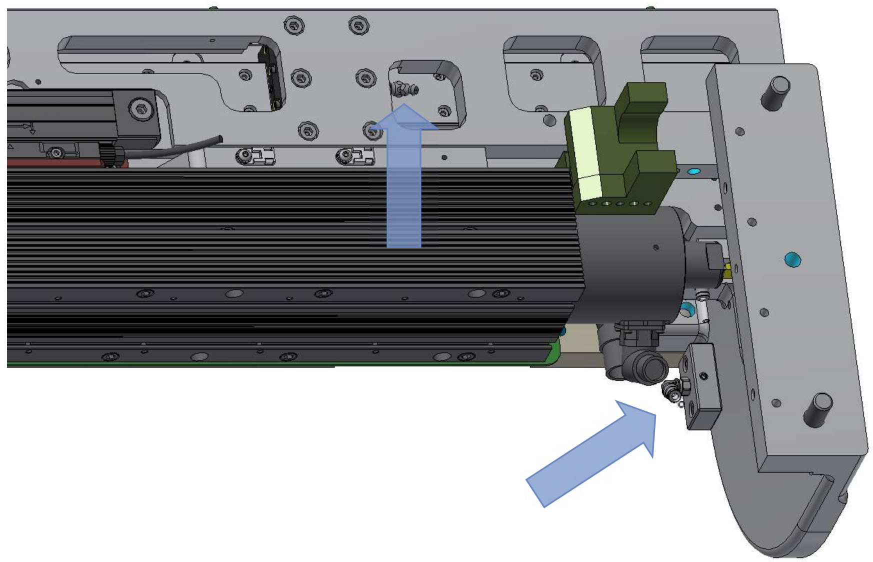
Biannual Maintenance (2x per year) or every 10 million cycles

The following task should be performed:

- Lubrication of the transfer & indexing system

Biannual Maintenance (2x per year) or every 10 million cycles

Linear Bearing Lubrication



- Lubricate pallet transfer & locking slides using the greasing nipples in the lower front of the machine on both sides.
- Grease & wipe off any surplus of grease.
- See the greasing chart.

4

Yearly

or 20 Million Cycles

Preventive Maintenance Mikron G05

Annual Maintenance or every 20 million cycles

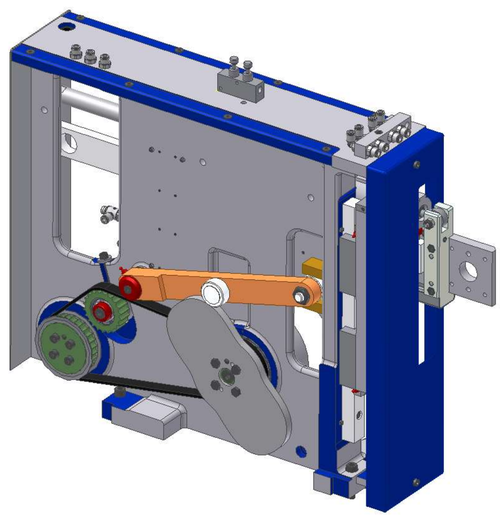
Once per year the following tasks should be performed:

- Check & lubrication of all the handling units
- Check the handling unit belts
- Check the control units
- Change the main gearbox oil
- Clean or change the electrical cabinet filters

Annual Maintenance or every 20 million Cycles

Pick and Place / Vertical Motion Unit Maintenance

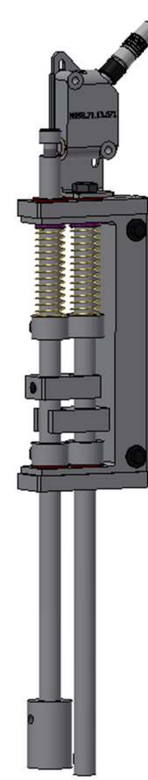
- Do the 5 million cycle maintenance.
- Lubricate linear guides and pivoting point using greasing nipples.
- Check that the rollers turn freely, if required replace them.
- Check driving belt condition & tension, if required replace them.
- See the greasing chart.
- See driving belt tension chart.



Annual Maintenance or every 20 million Cycles

Control and Inspection Units

- Inspect and clean the control units.

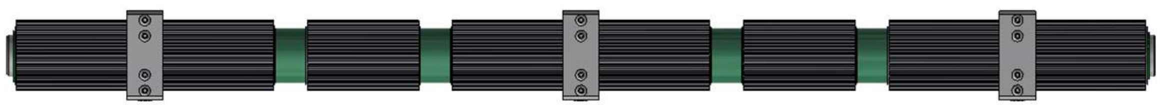


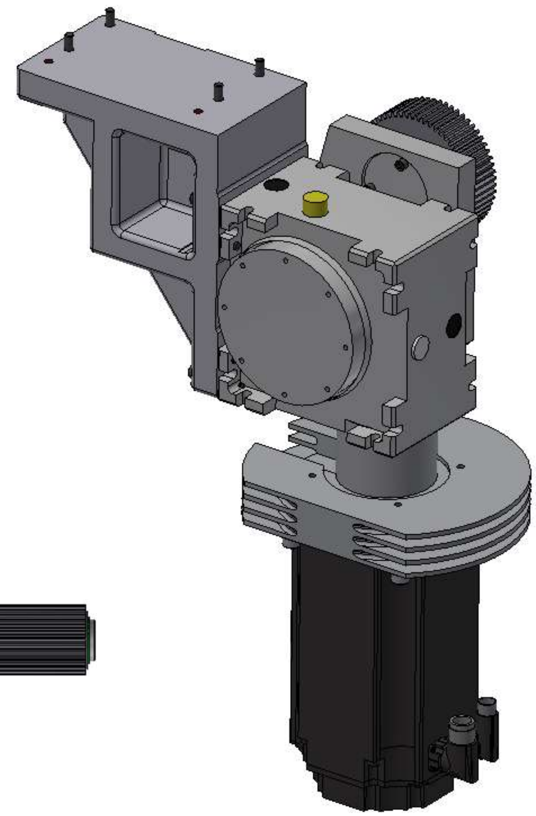
Annual Maintenance or every 20 million Cycles

Central Shaft

- Change the main gearbox oil.
- Gearbox oil capacity is approx.
  - Older gearboxes ≈ 0,33 liter
  - SGH 47 ≈ 0,5 liter
  - SGH 63 ≈ 0,8 liter

- Clean, check and grease the main shaft drive’s gears.
- See greasing chart.





Annual Maintenance or every 20 million Cycles

Electrical Cabinet Air Filtration

- Clean or change the filters on the electrical cabinet.
- Size of the filter: 170 x 170mm, 15mm thick.

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Lubrication and Tension

Quick Reference Guide

Lubrication Chart

WHAT	WHERE USED	MANUFACTURER / TYPE	MIKRON PART #
GREASE	CAMS, PUSHERS,ROLLERS	KLUBER KLUBERPASTE UH1 84-201	N928.43.13.006
GREASE	BALL BEARINGS, LINEAR BEARINGS, LINMOT SLIDER ROD	KLUBER KLUBERSYNTH UH1-14-31	N928.43.13.032
GREASE	GEARS, SPRINGS, GREASE ZERK FITTINGS	KLUBER KLUBERSYNTH UH1-14-151	N928.43.13.030 (1kg) N928.43.13.031 (400g)
OIL	CONTROL RODS, BALL JOINTS	INTERFLON FIN FOOD LUBE	N928.43.13.010
OIL	CENTRAL DRIVE GEARBOX	OLYT OLYT 3233	N928.43.13.013

Lubrication Devices

	Description	Grease type	Volume per block [cm³]	Interval [cycles] or [time]
P&P 120	Geared wheel (via deported lubrication point on central shaft)	UH1 14-151	0.1	5 mio or 3 months
	H lever cam follower (via deported lubrication point)	UH1 84-201	0.1	5 mio or 3 months
	H lever contact roller (via deported lubrication point)	UH1 84-201	0.1	5 mio or 3 months
	V lever cam follower (via deported lubrication point)	UH1 84-201	0.1	5 mio or 3 months
	V lever contact roller (via deported lubrication point)	UH1 84-201	0.1	5 mio or 3 months
	V lever pivot needle bearing	UH1 14-151	0.1	20 mio or 1 year
	H linear guide (front block)	UH1 14-31	0.3	20 mio or 1 year
	H linear guide (central block)	UH1 14-31	0.3	20 mio or 1 year
	H linear guide (rear block)	UH1 14-31	0.3	20 mio or 1 year
	V linear guide (upper block)	UH1 14-31	0.6	20 mio or 1 year
	V linear guide (lower block)	UH1 14-31	0.6	20 mio or 1 year
MV 120	Geared wheel (via deported lubrication point on central shaft)	UH1 14-151	0.1	5 mio or 3 months
	V lever cam follower (via deported lubrication point)	UH1 84-201	0.1	5 mio or 3 months
	V lever contact roller (via deported lubrication point)	UH1 84-201	0.1	5 mio or 3 months
	V lever pivot needle bearing	UH1 14-151	0.1	20 mio or 1 year
	V linear guide (upper block)	UH1 14-31	2.0	20 mio or 1 year
	V linear guide (lower block) - only for "2-blocks" version	UH1 14-31	2.0	20 mio or 1 year
Transfer & Indexing system	Geared wheels, left & right (via deported lubrication points on central shaft)	UH1 14-151	0.1	5 mio or 3 months
	Indexing lever cam follower (left & right) (via deported lubrication points)	UH1 84-201	0.1	5 mio or 3 months
	Indexing lever contact roller (left & right) (via deported lubrication points)	UH1 84-201	0.1	5 mio or 3 months
	Rise & Fall lever cam follower (left & right) (via deported lubrication points)	UH1 84-201	0.1	5 mio or 3 months
	Rise & Fall lever contact roller (left & right) (via deported lubrication points)	UH1 84-201	0.1	5 mio or 3 months
	Transfer V linear bearing (left & right) (via deported lubrication points)	UH1 14-31	2.0	20 mio or 1 year
	Transfer H linear bearings (left & right) (via deported lubrication points)	UH1 14-31	0.6	20 mio or 1 year
	Rise & Fall linear bearings (left & right) (via deported lubrication points)	UH1 14-31	0.3	20 mio or 1 year
	Transfer LinMot motor (lubrication via reservoir)	UH1 14-31	Thin film on slider after lubrication	20 mio or 1 year (inspect. every 3 months)

Belt Frequency Chart

Unit Type	Belt weight	Width	Belt span	Setup tension	Setup frequency (±5Hz)
PP 120* (1:1) & (1:2) **PolyChain** **& PolyChain-CARBON** **& Synchrochain CARBON**	4.7 kg/m^2	15.0 mm	237 mm (1:1)	New Belt 310 N	New Belt **140 Hz** (1:1) & (1:2)
			236 mm (1:2)	After Min 100h 229 N	After Min 100h **120 Hz** (1:1) & (1:2)
MV 120 (1:1) **PowerGrip**	5.8 kg/m^2	20.0 mm	275 mm	195 N	**75 Hz**
PP 60 (1:1) **PowerGrip**	5.8 kg/m^2	12.0 mm	257 mm	146 N	**89 Hz**
MV 60 (1:1) **PowerGrip**	5.8 kg/m^2	12.0 mm	289 mm	146 N	**79 Hz**
MV 120 (1:2) **PowerGrip**	5.8 kg/m^2	20.0 mm	272 mm	195 N	**75 Hz**
PP 60 (1:2) **PowerGrip**	5.8 kg/m^2	12.0 mm	255 mm	146 N	**90 Hz**
MV 60 (1:2) **PowerGrip**	5.8 kg/m^2	12.0 mm	286 mm	146 N	**80 Hz**
Not-Standard Belt (20mm CARBON)					
PP 120* (1:1) **PolyChain-CARBON** **& Synchrochain CARBON**	4.7 kg/m^2	20.0 mm	237 mm	New Belt 310 N	New Belt **121 Hz** (1:1) & (1:2)
				After Min 100h 229 N	After Min 100h **104 Hz** (1:1) & (1:2)

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Thank you

Making complexity accessible

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